



Hook

**KEA
DHCP**

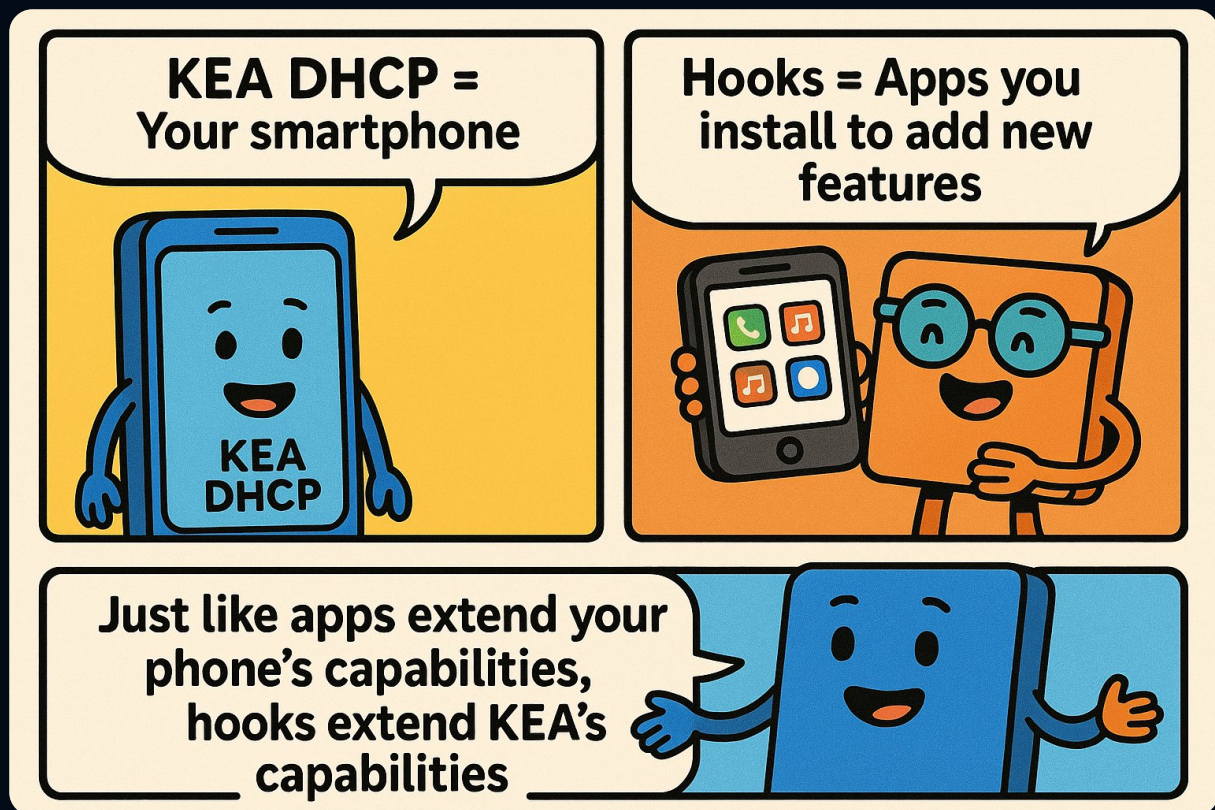
KEA DHCP HOOKS - 1ST GUIDE

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WHAT ARE KEA HOOKS ?

Think of KEA DHCP Hooks as plugins or extensions that allow you to customize and extend your DHCP server's behavior **WITHOUT** modifying the core KEA code.

Simple Analogy:



WHY DO WE NEED HOOKS ?

Traditional DHCP servers are pretty basic - they just hand out IP addresses. But modern networks need more:

- **Logging** to external databases
- **Authentication** with Active Directory
- **Dynamic DNS** updates
- **Custom lease management**
- **Integration** with network monitoring tools
- **Load balancing** across multiple DHCP servers

Hooks let you add these features without rebuilding KEA from scratch!

HOW DO HOOKS WORK?

KEA has hook points – specific moments in the DHCP process where it says “HEY, anyone want to do something here?”

MAIN HOOK POINTS:

1. **subnet4_select** - When choosing which subnet to use
2. **lease4_select** - When selecting a specific lease
3. **lease4_renew** - When a client renews their lease
4. **lease4_release** - When a client releases their IP
5. **pkt4_receive** - When any packet is received
6. **pkt4_send** - When any packet is sent

TYPES OF KEA HOOKS

1. LOGGING HOOKS

```
"hooks-libraries": [  
  {  
    "library": "/usr/lib/x86_64-linux-  
gnu/kea/hooks/libdhcp_mysql.so"  
  }  
]
```

- Logs all DHCP activity to MySQL/PostgreSQL
- Perfect for compliance and troubleshooting

2. HIGH AVAILABILITY (HA) HOOK

```
"hooks-libraries": [  
  {  
    "library": "/usr/lib/x86_64-linux-  
gnu/kea/hooks/libdhcp_ha.so",  
  
    "parameters": {
```

```
"high-availability": [  
  {  
    "this-server-name": "server1",  
    "mode": "hot-standby",  
  
    "peers": [  
      {  
        "name": "server2",  
        "url": "http://192.168.1.100:8000/",  
        "role": "standby"  
      }  
    ]  
  }  
]  
}
```

- Creates redundant DHCP servers
- Automatic failover if primary dies

3. RADIUS HOOK

```
"hooks-libraries": [  
  {  
    "library": "/usr/lib/x86_64-linux-  
gnu/kea/hooks/libdhcp_radius.so",  
    "parameters": {  
      "access": {  
        "servers": [  
          {  
            "hostname": "radius-server.company.com",  
            "port": 1812,  
            "secret": "your-radius-secret"  
          }  
        ]  
      }  
    }  
  }  
]
```

- Authenticates clients against RADIUS server
- Perfect for enterprise environments

4. FORENSIC LOGGING HOOK

```
"hooks-libraries": [  
  {  
    "library": "/usr/lib/x86_64-linux-  
gnu/kea/hooks/libdhcp_flex_id.so"  
  }  
]
```

- Creates unique identifiers for clients
- Useful for security and tracking

EXAMPLE: SETTING UP MYSQL LOGGING

Here's a practical example that logs everything to MySQL:

```
{
  "Dhcp4": {
    "interfaces-config": {
      "interfaces": ["ens33"]
    },
    "lease-database": {
      "type": "mysql",
      "host": "localhost",
      "name": "kea_db",
      "user": "kea_user",
      "password": "kea_password"
    },
    "hooks-libraries": [
      {
        "library": "/usr/lib/x86_64-linux-
gnu/kea/hooks/libdhcp_mysql.so"
      }
    ],
    "subnet4": [
      {
```

```
"subnet": "192.168.50.0/24",  
"pools": [  
  {  
    "pool": "192.168.50.100 - 192.168.50.200"  
  }  
]  
}  
]  
}  
}
```

POPULAR KEA HOOKS YOU SHOULD KNOW

Hook Name	What It Does	When to Use
libdhcp_mysql.so	MySQL database integration	Production logging, lease tracking
libdhcp_pgsql.so	PostgreSQL database integration	Enterprise environments
libdhcp_ha.so	High Availability	Mission-critical networks
libdhcp_radius.so	RADIUS authentication	Corporate networks with existing RADIUS
libdhcp_flex_id.so	Flexible client identification	Complex client identification needs

libdhcp_stat_cmds.so **Statistics and
monitoring**

Network monitoring
and analytics

HOOK CONFIGURATION BEST PRACTICES

1. ORDER MATTERS

```
{  
  "hooks-libraries": [  
    {  
      "library": "/path/to/first-hook.so"  
    },  
    {  
      "library": "/path/to/second-hook.so"  
    }  
  ]  
}
```

- Hooks are processed in the order listed!

2. ERROR HANDLING

Always check KEA logs when adding hooks:

```
sudo tail -f /var/log/kea/kea-dhcp4.log
```

3. PERFORMANCE IMPACT

More hooks = slower processing. Only use what you need!

KEA hooks transform a basic DHCP server into an enterprise-grade network service. They're modular, powerful, and let you customize DHCP behavior without touching the core code.

KEY TAKEAWAYS:

- ✓ Hooks = Plugins that extend KEA functionality
- ✓ They plug into specific points in the DHCP process
- ✓ Essential for enterprise environments
- ✓ Easy to configure once you understand the concept

QUICK REFERENCE COMMANDS

Install KEA with hook libraries (Updated 2024+ method)

```
sudo apt install isc-kea
```

Alternative install specific components

```
sudo apt install isc-kea-dhcp4-server isc-kea-admin
```

Check available hooks

```
ls /usr/lib/kea/hooks/
```

Validate KEA Configuration

```
sudo kea-dhcp4 -t /etc/kea/kea-dhcp4.conf
```

Start KEA DHCP service

```
sudo systemctl start isc-kea-dhcp4-server.service
```

Monitor Logs

```
sudo journalctl -u isc-dhcp4-dhcp-server -f
```